Systems of Linear Equations

30 problems · A worked example comes before each type. Show every step.

Name: _____ Date: _____

Key Idea

A **system** is two equations at the same time. Its **solution** is the point (x, y) that makes **both** equations true — where the two lines cross.

- **Graphing**: draw both lines and read the crossing point.
- **Substitution**: one equation already says "$y = \ldots$" (or $x = \ldots$)? Put it into the other.
- **Elimination**: add or subtract the equations so one letter disappears.

Common Mistake

A solution must work in **both** equations — always check in the one you did *not* use last.
When you subtract an equation, subtract *every* term: $(3x + 2y) - (x - 2y) = 2x + 4y$, **not** $2x$.

Type 1 Is It a Solution?

Watch Me — Worked Example

Is $(4, 1)$ a solution of $x + y = 5$ and $x - y = 3$?
Put $x = 4$, $y = 1$ into **both** equations.

Equation 1 $4 + 1 = 5$ ✓

Equation 2 $4 - 1 = 3$ ✓

Answer **Yes** — it works in both.

1. Is $(2, 3)$ a solution of $x + y = 5$ and $2x - y = 1$?

Yes / No: _____

2. Is $(1, 4)$ a solution of $y = 3x + 1$ and $x + y = 6$?

Yes / No: _____

3. Is $(-1, 2)$ a solution of $2x + 3y = 4$ and $y = -2x$?

Yes / No: _____

4. Is $(3, -2)$ a solution of $x - y = 5$ and $x + 2y = 1$?

Yes / No: _____

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Type 2 Solve by Graphing

Watch Me — Worked Example

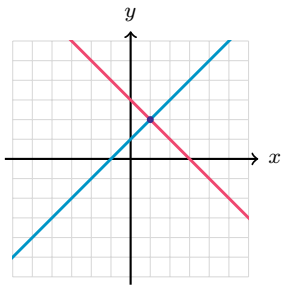
Solve by graphing: $y = x + 1$ **and** $y = -x + 3$**.**

Step 1 Graph $y = x + 1$: start at $(0, 1)$, up 1, right 1.

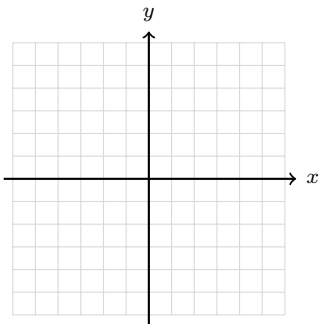
Step 2 Graph $y = -x + 3$: start at $(0, 3)$, down 1, right 1.

Step 3 They cross at $(1, 2)$. Check: $2 = 1 + 1,\ 2 = -1 + 3$.

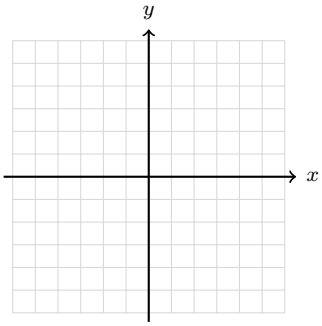
Answer $\mathbf{(1, 2)}$



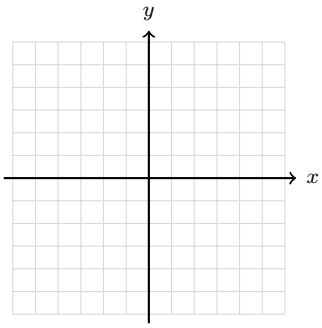
5. Solve by graphing: $y = x + 1$ and $y = -x + 5$.



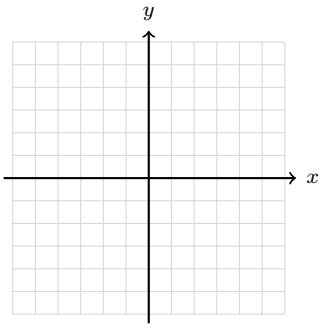
6. Solve by graphing: $y = \frac{1}{2}x - 2$ and $y = -x + 4$.



7. Solve by graphing: $y = 2x - 3$ and $y = -x + 3$.



8. Solve by graphing: $y = -x + 1$ and $y = x - 3$.



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Type 3 Substitution

Watch Me — Worked Example

Solve: $y = 3x$ **and** $x + y = 8$**.**

The first equation tells you what y is. Replace y in the second equation with $3x$.

Step 1 Substitute $x + 3x = 8$

Step 2 Solve $4x = 8,\ x = 2$

Step 3 Find y $y = 3 \cdot 2 = 6$

Check $2 + 6 = 8$ ✓

Answer $\mathbf{(2, 6)}$

9. Solve: $y = 2x$ and $x + y = 12$

$(x, y) =$ _____

10. Solve: $y = x + 3$ and $2x + y = 9$

$(x, y) =$ _____

11. Solve: $x = y - 1$ and $3x + y = 13$

$(x, y) =$ _____

12. Solve: $y = -3x + 2$ and $x - y = 6$

$(x, y) =$ _____

13. Solve: $y = 4x - 5$ and $y = 2x + 1$

$(x, y) =$ _____

14. Solve: $x + 2y = 7$ and $x = 3y - 8$

$(x, y) =$ _____

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Type 4 Elimination

Watch Me — Worked Example

Solve: $3x + 2y = 16$ **and** $x - 2y = 0$**.**

The y-terms are $+2y$ and $-2y$: adding the equations makes y disappear.

Step 1 Add the equations	$4x = 16$
Step 2 Solve	$x = 4$
Step 3 Find y	$4 - 2y = 0,\ y = 2$
Check	$3 \cdot 4 + 2 \cdot 2 = 16$ ✓
Answer	$\mathbf{(4, 2)}$

Notice

No opposite terms? **Multiply first.** In $x + 2y = 8$ and $3x + 4y = 20$, multiply the first equation by 2 so both have $4y$, then subtract.

15. Solve: $x + y = 10$ and $x - y = 4$

$(x, y) =$ _____

16. Solve: $2x + 3y = 13$ and $2x - y = 1$

$(x, y) =$ _____

17. Solve: $3x + y = 11$ and $x - y = 1$

$(x, y) =$ _____

18. Solve: $x + 2y = 8$ and $3x + 4y = 20$

$(x, y) =$ _____

19. Solve: $2x + y = 7$ and $3x - 2y = 7$

$(x, y) =$ _____

20. Solve: $4x + 3y = 1$ and $2x - y = 3$

$(x, y) =$ _____

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Type 5 How Many Solutions?

Watch Me — Worked Example

Compare slopes and y-intercepts — no need to solve.

Slopes	y-intercepts	Solutions
different	—	**one** (the lines cross)
same	different	**none** (parallel lines)
same	same	**infinitely many** (the same line)

Example: $y = 2x + 3$ and $y = 2x - 1$: same slope 2, different intercepts $\Rightarrow$ **no solution**.

21. How many solutions?
$y = 2x + 1$ and $y = 2x - 4$

Answer: _____

22. How many solutions?
$y = 3x - 2$ and $y = -x + 6$

Answer: _____

23. How many solutions?
$y = \frac{1}{2}x + 3$ and $2y = x + 6$

Answer: _____

24. How many solutions?
$x + y = 5$ and $x + y = 9$

Answer: _____

25. How many solutions?
$3x - y = 2$ and $6x - 2y = 4$

Answer: _____

26. How many solutions?
$y = -4x + 1$ and $y = 4x + 1$

Answer: _____

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Type 6 Word Problems

Watch Me — Worked Example

Adult tickets cost $8 and child tickets cost $5. A family buys 6 tickets for $39. How many of each?

Two unknowns, two facts: one equation for **how many**, one for **how much money**. Let a = adults, c = children.

Step 1 How many	$a + c = 6$
Step 2 How much	$8a + 5c = 39$
Step 3 Substitute $c = 6 - a$	$8a + 30 - 5a = 39,\ 3a = 9,\ a = 3$
Answer	**3 adult and 3 child tickets** (check: $24 + 15 = 39$)

27. The sum of two numbers is 20 and their difference is 6. Find the numbers.

Answer: _____

28. A farm has chickens and cows: 15 heads and 44 legs. How many of each?

Answer: _____

29. Pens cost $2 and notebooks cost $3. Leo buys 10 items for $24. How many notebooks?

Answer: _____

30. Gym A charges $30 to join plus $10 a month. Gym B charges $15 to join plus $15 a month. After how many months do they cost the same? What is that cost?

Answer: _____

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